

A MOMENT THAT CAN'T WAIT

It's 11 PM.

Your kid just

cut their hand.

You're out of bandages.

45 min instant delivery.

And the **delivery fee?**

More than the bandages.

45 min delivery window.

\$10 minimum basket + delivery fee.

Just for a bandaid?

294M

ORDERS / YEAR

~45 min

AVG TODAY
FOR INSTANT
DELIVERY

\$104

Instacart's
average order
value

Sources: Maplebear Inc. (DBA Instacart) SEC Form 8-K, Q4 2024. Delivery time based on Instacart's advertised instant delivery windows.

THE OPTIONS RIGHT NOW



Call an ambulance

A minor cut — completely overkill

Overkill



Drive yourself

Leave a distressed child home alone at 11 PM

Not an option



Instacart today

45 min delivery. Fee costs more than the item.

45 min

Too slow

Ambulance
overkill

OPPORTUNITY



the gap

Instant delivery
today
45–60 min

What if there was a **better way?** For something that **doesn't already happen?**

*Average order Value more than \$100!
Alas, urgencies can often come in **small packages!***

Speed isn't a feature — it's **the promise.**

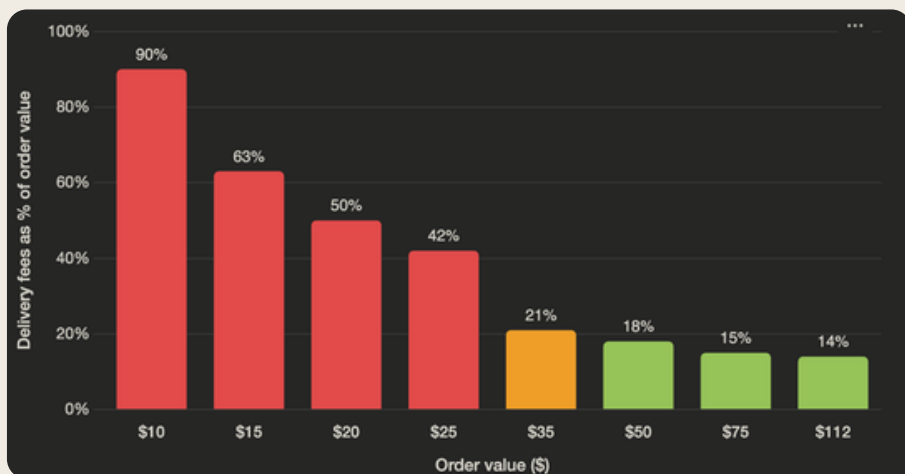
BREAKING DOWN THE PROBLEM- ONE MILE AT A TIME!

Shoppers get it to the door.
But the door is getting *further away*.
And none want to drive just for a bandaid!

Instacart's mission is to make grocery/essentials accessible to **every household**.

Today, the platform reaches **85% of U.S. homes** across **14.4M active users** - which implies huge penetration.

Delivery fees as % of Order value



And high delivery fees is what keeps the users from making small purchases! **Market Opportunity!**

LANDSCAPE



Grubhub + Avride

Sidewalk delivery robots. Expanding from university campuses to public markets.
Jersey City



Walmart + Wing

270+ stores, 8 metros by 2027. 150K+ deliveries, <19 min avg.
DFW, Atlanta, Orlando, Tampa, Charlotte



Zipline + Chipotle

"Zipotle" drone delivery. FAA Part 135, BVLOS certified. ~5 lb payload.
Rowlett TX · expanding DFW metro



DoorDash Dot

In-house robot via DoorDash Labs. Building dispatch + routing for last-mile.
Phoenix metro



Amazon Prime Air

MK30 drone, sub-60 min. Vertically integrated fulfillment + logistics.
College Station TX, Lockeford CA



Uber Eats + Serve Robotics

Sidewalk robots via Uber Eats. Deploying thousands across U.S. cities.
LA, Dallas, Miami, Atlanta · expanding nationally

TAM — GLOBAL
LAST-MILE
DELIVERY

\$186B

SAM — US share:
29% of Global

\$33– 37B

SOM — 10% of SOM

\$3B

ORDERS THAT DON'T HAPPEN TODAY

Autonomy doesn't replace shoppers — it **unlocks** demand that doesn't exist yet



THE OPPORTUNITY

Autonomous delivery **bridges the gap** between vision and today's constraints

- Vision alignment:** Instant, affordable delivery everywhere
- Constraint:** As delivery distance grows, the model strains. ETAs stretch, costs rise, and shoppers skip batches that don't pay.
- Unlock:** Drones & robots enable sub-20 min at lower cost

WHY INVEST NOW

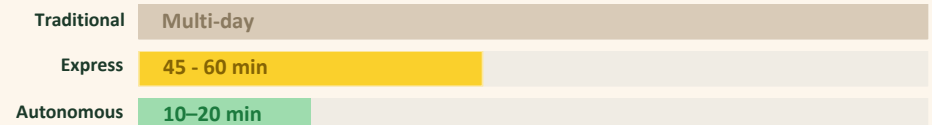
One autonomous layer scales across 1,800+ retail partners, a moat no single-brand competitor can replicate

Feasible with minimal changes

Market is Moving in this direction!

Regulatory window: Drone and robotics companies receiving clearances across cities

Demand proven: 150K+ Walmart deliveries; consumer appetite is real



EXPERIENCE EVOLUTION

How each stakeholder evolves!

STAKEHOLDER	Today Pain points	With autonomous Evolution	Risks & mitigations Guardrails
IC Instacart	Order → Find shopper → Shop+deliver → 45-60 min / multiday \$10-12/delivery Shopper tied up 45-60 min Suburban gaps	Order → Auto-route → Drone/robot dispatch → 15-20 min \$5-12 at scale "Express" +\$7/order 85%→95%+ coverage	Low margins <\$10 baskets High robot throughput (3-4/hr); ad surfaces generate retail media revenue. Weather grounds drones Auto-reroute to robots in rain/wind. Dual-mode redundancy.
SH Shoppers	Accept → Drive to store → Pick → Drive far → Hand off 30-45 min driving/order Long trips in bad weather Fuel + wear costs Tips-dependent pay	Placed in-store → Pick + stage → Load vehicle → No long drives No far/unsafe trips Save time + fuel +20-30% picks/hr Can still deliver if needed	Backlash at scale Autonomous handles trips shoppers don't want — not replacement. More volume = more batches. Tip earnings loss Dedicated stager/picker role with hourly minimums. Shoppers still deliver large/complex orders that fetch high tip.
RT Retailers	Shopper enters → Walks aisles → Checkout → Drives away Unpredictable traffic Limited to driver radius ~2K SKUs via Express	Staff picks → Dispatch lane → Auto-loaded → Dispatched 2-3x order frequency Radius 2-3x (drones) Top 100 SKU drone catalog	Retailer resistance Start with deepest Instacart-integrated partner, prove ROI, then expand. Staging infra needed Dedicated dispatch lanes; use existing back-of-store areas. Minimal layout change.
CX Consumers	Browse → Checkout → Wait 30-60 min → Doorbell \$4-10 fee No urgency option Suburbs underserved	Browse → "Express" → Live tracking → Curb drop ~15 min 10-20 min delivery \$2-5 savings (robot) OTP / secure drop-off	Customer trust Real-time tracking, tamper-proof compartments, photo confirmation. Theft or tampering Locked compartments, PIN-code unlock, GPS alerts.

Every stakeholder gains. Shoppers pick more, retailers sell more, customers get faster, Instacart earns more.

● THE PILOT - DRONE

Plano, TX: Drone delivery from **ALDI - LOW HANGING FRUIT**

Dallas Fort Worth region is the **most advanced drone delivery metro** in the U.S. We pilot in Plano for it's **open airspace, stable weather, dense single-family homes, and high Instacart demand.**

Why Plano, Texas

- **Suburban airspace** — clear of Love Field & DFW International
- **Stable weather** — consistent flying conditions year round
- **Open drop zones** — yards & driveways for safe landing
- **High demand** — Frisco, Prosper, Allen, Plano in range



Why ALDI - 8801 Stacy Rd, McKinney, TX 75070

- **Deepest Instacart integration** — powers ALDI's entire U.S. e-commerce
- **ALDI Express** — ~2,000 curated SKUs, 30 min delivery
- **Minimal infra changes** — Partner Pick + loading zone
- **Clean data** — single store, isolates delivery learnings
- **Diverse Inventory**

● DRONE PARTNER & COVERAGE



Zipline Platform 2

- **Already in DFW** — delivering for Walmart & Chipotle (Zipotle)
 - **FAA regulations** — cleared for commercial drone flights in DFW
 - **Tethered droid** — hovers at ~300 ft, lowers within ~1 m precision
- 5 lb max · ~65 mph · 5–8 mi range · 10–20 min



● THE PILOT - SIDEWALK ROBOT

Dallas, TX: Robot delivery from ALDI - LOW HANGING FRUIT

Dense urban delivery where drones can't fly and shoppers are scarce. Robots own the gaps, **no-fly zones, late night, bad weather, peak hour**. And delivery cost, haha!



Why Dallas, Texas - 4120 Gaston Ave, Dallas, TX 75246

- **No-fly zone** — drones restricted, robots are the only autonomous option. Demand outruns shoppers in peak hours.
- **Dense sidewalks** — East Dallas, Lakewood, Lower Greenville — continuous paths
- **Apartment clusters** — front-door delivery where drones have no drop zone
- **High demand** — Dense Downtown



Why two locations of one retailer - ALDI?

- **Same SKU Catalog**
- **Different Geography** — Through testing of autonomous delivery
- **Isolate Delivery Learnings** — single retailer means any difference is the delivery mode, not the store
- **One Integration**

● Robot PARTNER & COVERAGE



Serve Robotics

- **Already in DFW** — delivering via Uber Eats in Uptown Dallas
- **Level 4 autonomy** — AI-powered navigation, no remote operator needed
- **Insulated compartment** — contactless, tamper-proof delivery to the door
25 lb max · 3–4 mph · 1–2 mi range · 20–30 min

The Gen3: Our Fastest, Smartest, Most Reliable Robot Yet

Doubling Delivery Speed, Boosting Range, and Slashing Costs by 50%

5x more brain power with Nvidia Jetson Orin and Ouster REV7 Lidar for ultra-fast decisions

Advanced collision avoidance and enhanced emergency braking for next-level safety

40% faster stops for quick reaction times, even in busy environments

Extended 14-hour operations with 67% more battery capacity and 15% more cargo space

60% faster top speeds and a smoother ride

Built tough to handle diverse weather and terrain conditions

50% reduced cost, providing faster ROI



FEATURES

- 1 Full-stack AI sensors; Level 4 autonomy
- 2 Touch-screen interface
- 3 13 gallons of cargo; Holds up to 4x 16" pizzas
- 4 Redundant connectivity
- 5 11 mph top speed; All-terrain drivetrain
- 6 All day battery



	GEN2	GEN3
Top Speed	7 mph (2.8 m/s)	11 mph (4.9 m/s)
Weather	32 to 104°F / Light Rain	-4 to 113°F / Heavy Rain
Range	23 miles (10 hours)	48 miles (14 hours)



● THE PILOT - COST CURVE AND BREAK EVEN ANALYSIS

Automation gets cheaper with scale.

Per-delivery cost (what Instacart pays the partner)

Drone (Zipline)	\$14–20 per delivery	<i>Sacra est.</i>
Robot (Serve)	\$3–7 per delivery	<i>Serve 10-K derived</i>
Shopper (today)	\$7–12 per delivery	<i>industry known</i>
At scale (projected)	\$3–5 drone · \$3–4 robot	<i>industry benchmarks</i>

Volume & revenue assumptions

Pilot period	3 × 6-month phases (18mo total)
Orders/day ramp	200 → 800 → 1,600
Avg basket value	\$9–11 (grows with trust)
Customer delivery fee	Free (IC+) · \$0.99–1.99 (non-sub)

3 phased 18 month MVP: Launch small. See what works. Prove economics before expanding nationally.

Fleet for each month (owned by Zipline/Serve, not Instacart)

Drones	6 → 18 → 36 (across 1 → 3 → 6 nests)
Robots	15 → 60 → 120 (across 1 → 3 → 5 depots)

Instacart's one-time investment (all estimates)

API + routing + QA + marketing + insurance
Zero fleet capex · zero ops team · zero new headcount
No public source for integration costs — author's estimates

\$220–340K

Disclaimer: All data from public sources. Partner fees and projections are estimates — Zipline/Serve don't disclose B2B terms. Financial model is directional, not a forecast. Many values shown are estimates for representational purpose.

Sources: [Sacra](#) · [Serve 10-K](#) · [DroneXL](#) · [CNBC](#) · [Low Altitude Economy](#)

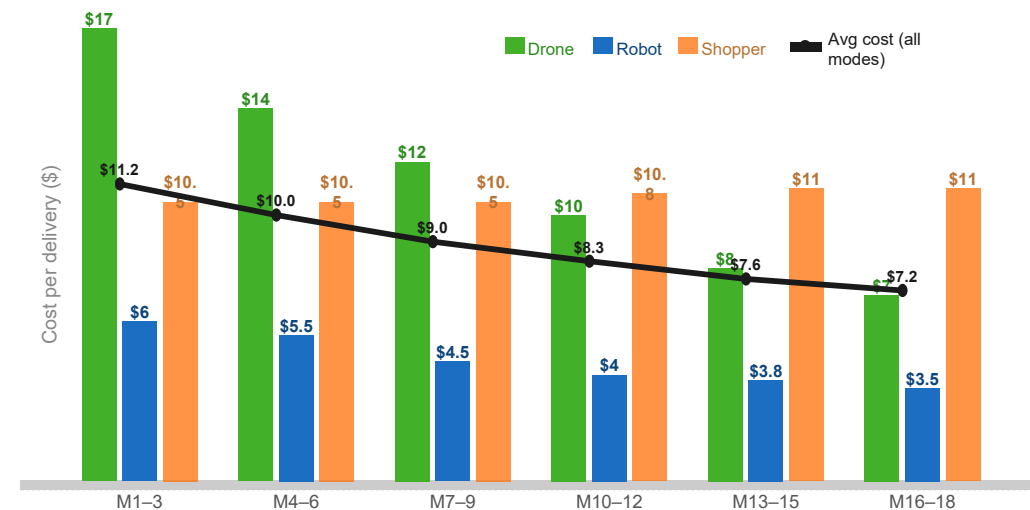
Revenue per period

= (avg delivery fee × orders/day × 180 days)
+ (basket margin 30% × avg basket × orders/day × 180 days)

Costs per period

= (blended partner fee × orders/day × 180 days) + Instacart overhead

	M1–6	M7–12	M13–18
Orders/day	200	800	1,600
Revenue	\$270K	\$1.19M	\$2.98M
Costs	\$344K	\$890K	\$1.41M
Profit	–\$74K	+\$297K	+\$1.57M
Cumulative	–\$74K	+\$223K	+\$1.79M
Break-even: Month 8 (est.) against investment of \$220 - \$340k			



High fixed expenses for automation are **spread over high delivery volumes**, whereas manual costs remain flat due to persistent human labor needs.

MVP FEATURES TO BUILD

FOR CUSTOMER

- Autonomous eligibility indicator on cart
- Live tracking and amount saved
- Delivery spot selection
- Unlock code for robot compartment

FOR STORE ASSOCIATE / SHOPPER

- Instructions to stage or deliver manually
- Vehicle details (drone ID or robot ID)

FOR INSTACART PLATFORM

- Eligibility engine (weight, address, zone)
- Zipline + Serve API integration
- Geo-fence + auto-fallback to shopper
- Pilot analytics dashboard

METRICS FRAMEWORK

NORTH STAR

Incremental orders fulfilled via autonomous delivery

- Proves expansion, not cannibalization
- More orders = more revenue for all
- Measures if drones/robots are useful
- Retailer-friendly growth signal

SUPPORTING METRICS

CONSUMER

- Overall NPS (all modes)
- Repeat order rate
- Delivery damage / loss rate

STORE OPERATIONS

- Pick-to-stage time
- Staging error rate
- Associate satisfaction

ALDI (PILOT RETAILER)

- Incremental order volume
- Express SKU out-of-stock
- Avg basket size change

COUNTER-CANNIBALIZATION

- Non-ALDI retailer order volume change
- Shopper earnings delta in DFW
- Cross-retailer migration rate

INSTACART PLATFORM

- Autonomous dispatch success
- Cost per autonomous delivery
- Autonomous share of eligible
- Shopper fallback rate
- Net-new orders (causal lift)
- API uptime (Zipline + Serve)

Alarm Bell: Non-pilot retailer volume or shopper earnings drop >5% vs control → pause and investigate.

WHAT WE'RE HOPING TO LEARN

1 Incremental demand

Do autonomous modes unlock new urgent, low-basket orders?

→ Is this market expansion?

2 Routing effectiveness

Is the system assigning drone → urgent, robot → medium, shopper → complex correctly?

→ Can we trust auto allocation?

3 Speed vs cost

Do faster deliveries lift frequency and retention? Do lower-cost modes improve margins?

→ Optimal mode mix?

4 Operational feasibility

Can stores handle pick + stage? SLAs across range, no-fly zones, weather?

→ Can this scale?

5 System reliability

Autonomous success rate and fallback rate to shopper.

→ Consistently reliable?

6 Retailer value

Change in orders per store, basket size, order frequency.

→ Expand to other retailers?

7 Shopper role evolution

Can shoppers transition to picking + staging? Throughput impact?

→ Sustainable workforce model?

8 Counter-cannibalization

Are non-ALDI retailers losing share, or is total DFW pie growing?

→ Safe to scale marketplace-wide?

Thank you

Access the customer side prototype here (<https://rohanjauhari13.github.io/>)

Rohan Manish Jauhari
jauhari.r@northeastern.edu